

Development of Internet of Things concept for Precision Beekeeping

Aleksejs Zacepins, Armands Kviesis
Faculty of Information Technologies
Latvia University of Agriculture, Latvia
Corresponding e-mail: alzpostbox@gmail.com;

Aldis Pecka, Vitalijs Osadcuks
Faculty of Engineering
Latvia University of Agriculture, Latvia

Abstract— Implementation of precision apiculture methods into practice is highly depended on availability and cost-effectiveness of honey bee monitoring equipment. Expenses of monitoring system installation and maintenance activities also should be minimized. In order to successfully predict different bee colony states like: swarming, determine nectar collection activity and health of honey bees both in active and passive wintering period, minimum hive parameters that should be monitored are temperature and mass. This research presents conceptual design of Internet of Thing approach for the beekeeping. As well honey bee colony temperature monitoring system based on local wireless sensor network together with GSM/GPRS external interface for packet-based communication with remote server on the Internet is developed. Logical topology of the local network is star with one central module. The hive group consists of 10 units. Each hive in the group is equipped with integral temperature sensor chip ($\pm 0.4^{\circ}\text{C}$ accuracy), low power microcontroller and radio frequency (RF) transceiver module. All components are powered by alkaline cell battery, which can provide energy for one year minimum of autonomous operation with 10-minute data logging interval. The central module is also placed on one of the hives and consists of temperature sensor, RF and GSM/GPRS modules, photovoltaic cell array, battery, charge controller and minimal user interface for operational status reporting. The system allows sensor data logging to local storage and periodic sending to a remote server for further detailed analysis in user web or mobile application.

Keywords - Precision Apiculture; Precision Beekeeping; monitoring system, smart apiary, internet of things

I. INTRODUCTION

Beekeeping is important production branch of the agriculture. Honeybees are very valuable economical insects and they are the main pollinators in the world [1]. Usage of ICT in apiculture can increase the beekeepers' knowledge about behaviour of individual bee colonies and improve the efficiency of beekeeping by bringing it to the next technological level [2].

Nowadays Cloud Computing and The Internet of Things (IoT) are the two hot points in the Internet field [3]. The application of the two new and trending technologies is in hot discussion and research, but still quite less on the field of agriculture and especially in the beekeeping, however IoT has great potential in Agriculture. In this paper, authors describe developed honeybee colony temperature monitoring system within the IoT concept, using the wireless sensor networks.

In recent years, extensive attention has been paid to Internet of Things [4]. The IoT can be defined as a network of Internet enabled objects linked with web services that interact with these objects [5]. Recent advances in sensor, networking and RFID technologies allow connecting unlimited count of physical world objects to the global IT infrastructure, which could, ultimately, enable realization of the IoT [6]–[8]. IoT represents the future of computing and communications and its development depends on dynamic technical innovation in a number of important fields.

The basic idea of the IoT is that almost every physical thing in our world can also become a computer or a data generating device that is connected to the Internet. The IoT idea and concept is not new. However, it only recently became relevant to the practical implementations, mainly because of the huge progress made in hardware development. The decline of size, cost and energy consumption, hardware dimensions that are closely linked to each other, now allows the manufacturing of extremely small and inexpensive low-end devices and microcomputers [5].

IoT concept is very closely linked to wireless sensors and wireless sensor networks technology, because wireless devices are main actors for data collectors and transmitters. Wireless sensor network can be defined as a spatially distributed system of autonomous sensor nodes [9]. Wireless sensor networks can be applied almost in all branches. Sensors can be placed on stationary or on moving objects [10].

The IoT and wireless sensor network technologies can be used also in beekeeping for remote and real-time monitoring of honeybee colonies. This monitoring system can be considered as a part of the whole Precision Apiculture (Precision Beekeeping, PB) system. Precision Beekeeping is recently defined as a scientific direction where Information Technologies (IT) and beekeeping are linked together. PB is defined as an apiary management strategy based on monitoring of individual bee colonies to minimize the resource consumption and maximize the productivity of bees [11], [12]. Implementation of IoT and usage of wireless solutions in beekeeping still is not developed very well. There are very few examples of approaches and solutions: temperature and humidity controlling system, which is powered by solar energy [13]; open-source bee-hive monitoring system developed by Glyn and Clive Hudson [14]; project called "PIC'n the Beehive" [15].

The aim of this paper is to describe a developed IoT system prototype for honey bee colony temperature monitoring. Temperature monitoring of bee colonies is relatively cheap and easy way to have some crucial key information about behaviour and state of the colony [2].

II. SYSTEM COMPONENTS

As usually beekeepers place their colonies outside the living regions in rural areas (for example in open fields or even in forests). Advantage of authors proposed system is that it can be located in any place and there is no need for grid power supply, because it can be operated using batteries and solar power.

Authors proposed IoT concept for Precision Beekeeping consists of several key components: hardware part (sensor network with multiple wireless nodes and central module with WAN interface), software part and data analysis, which is integrated in software. To this moment wireless system is developed for monitoring of 10 bee colonies, but this number can be easily increased by simply adding additional nodes and theoretically there are no upper limit for the number of monitored objects. In design proposed by authors only one byte is used for device address, so this limits the maximum number to 254 devices, which is still enough, because usually one apiary consists of up to 50 colonies.

For every sensor network to be economically reasonable it is necessary to minimize cost price of each node as well as to reduce to minimum installation and maintenance efforts.

Block diagram of a wireless sensor node is shown in Fig. 1. For bee colony temperature measurements integral analog output sensor LMT85 is used. At required application temperature range (-10 to 40 °C) it performs with typical ± 0.4 °C and maximum error ± 1.8 °C (3.6% for selected range). Average accuracy is compensated with good output signal stability over operational power supply range – line regulation is $200 \mu V \cdot V^{-1}$, which according to sensor gain $8.2 mV \cdot ^\circ C^{-1}$ corresponds to 0.024 °C. Therefore static sensor error can be compensated in software. Such accuracy is adequate for the main task – swarming detection as only temperature differences by $1.5\text{-}3.4^\circ C$ [16]. LMT85 pricing for high volumes is less than 0.50 EUR per unit.

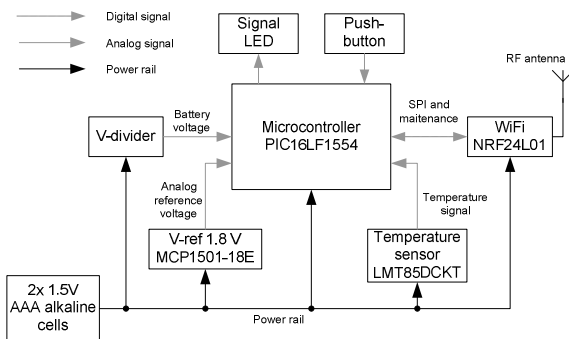


Figure 1. Block diagram of a wireless sensor node

Low power microcontroller unit (MCU) PIC16LF1554 acts as the main control component, which measures temperature and battery voltage, stores measurements in temporary memory and periodically (in authors case it is 10 min, but this can be easily changed to needed time interval) sends data to central module.

Two alkaline 1.5 V cells with typical capacity 860-1200 mAh (depending on discharge rate) are used as main power source. The cells are connected in series, this configuration will supply 3 V nominal down to 2 V, this fully conforms to power supply requirements of all sensor node components. Thus, there is no need of any additional voltage regulating elements. Drawback of this approach is necessity of using additional voltage reference for accurate analog measurements.

Communication is done using NRF24L01 chip based wireless module. It is very energy efficient (up to 12.3 mA in active receive and down to 900 nA in power down mode) operates on RF 2.4 GHz band physical and data link layer and incorporates propriety frame handling and retransmission protocol for reliable data exchange. Interface to microcontroller is SPI and 3 additional pins for control and interrupts.

LED and pushbutton are not needed during normal operation, but are used for simple user interface during sensor node and central module pairing procedure and.

The main responsibility of central module (Fig. 2) is to listen paired sensor nodes, collect temperature data and send it to cloud server using GPRS. It is also equipped with the same LMT85 sensor and voltage reference for local temperature measurements, status LEDs and A6 GSM/GPRS module. The central module is controlled by PIC24FV16KA304 microcontroller.

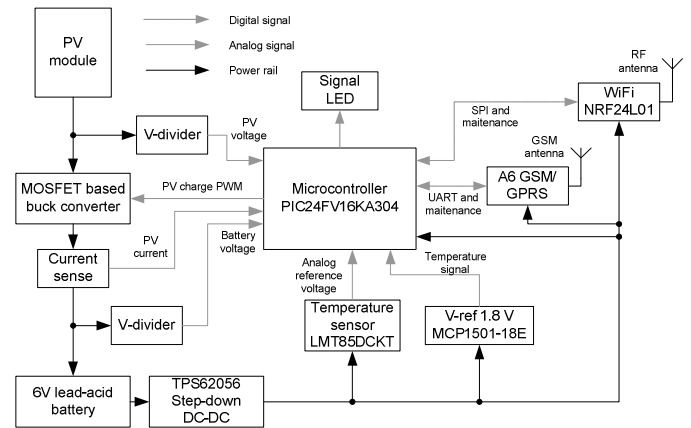


Figure 2. Block diagram of central module

In order to power more energy consuming components, unlike sensor nodes the central module is equipped with 6 V 1 Ah rechargeable sealed lead-acid battery as well as solar 12 V, 5 W PV module. This allows the device to be almost completely energetically self-sufficient. Lead-acid battery charging from PV power is performed using simple buck

converter with one inductor and power MOSFET. Converter is controlled by MCU both to maintain desirable charging voltage and current and to track optimum power point and maximize efficient energy usage from PV. Voltage feedbacks are through simple voltage dividers, battery charging current feedback is implemented using shunt and TS1100 current sense amplifier.

Fig. 3 shows the prototypes of both described devices (central module and sensor nodes).

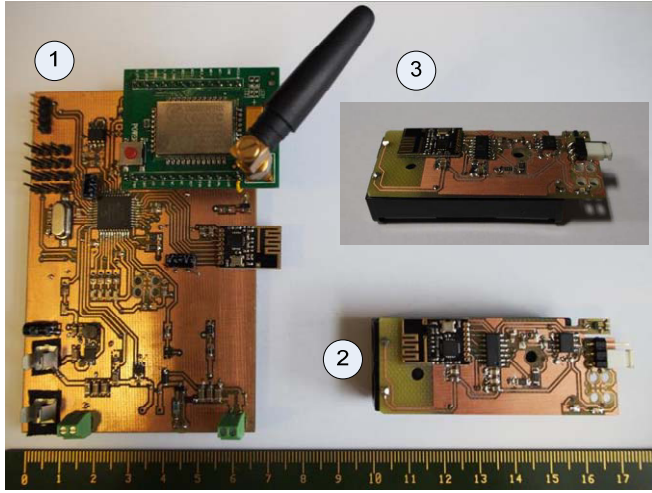


Figure 3. Hardware elements for monitoring system development: 1 – main module; 2 – sensor node; 3 – side view of sensor node with battery compartment

Power consumption tests were performed for sensor node as batteries is the only source available. Software of the sensor node MCU has been organized with heavy use of built-in energy conserving features. MCU is operating using its internal calibrated oscillator at 4 MHz, but most of the time MCU is held in sleep mode with nominal current consumption. Temperature and battery voltage measurement (for monitoring purposes) is performed once every minute, but data transmission every 10 min. Each transmission holds the newest minute-interval data from last successful transmission (Fig. 4).

Wireless sensor range test proved, that data can be reliably send up to a distance of about 20 m in open field.

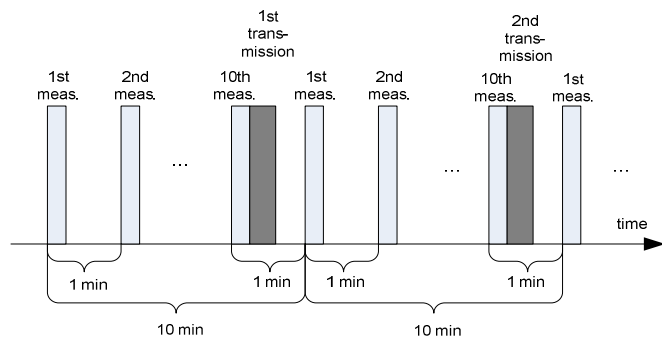


Figure 4. Sensor node operation sequence

Power consumption for analog temperature and battery voltage measurement and transmission process is depicted in Fig. 5 and Fig. 6. The graphs were captured in debug mode using Microchip RealICE emulator with PowerMonitor plug-in board. Average sensor board power consumption, when MCU is in sleep mode is 148 μ W (measured using two Fluke 189 multimeters).

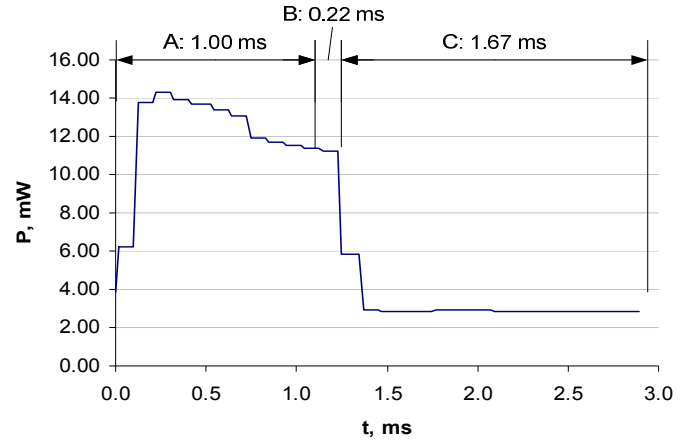


Figure 5. Power consumption during temperature and battery voltage measurement: A – voltage reference start-up; B – AD conversion; C – calculation of results

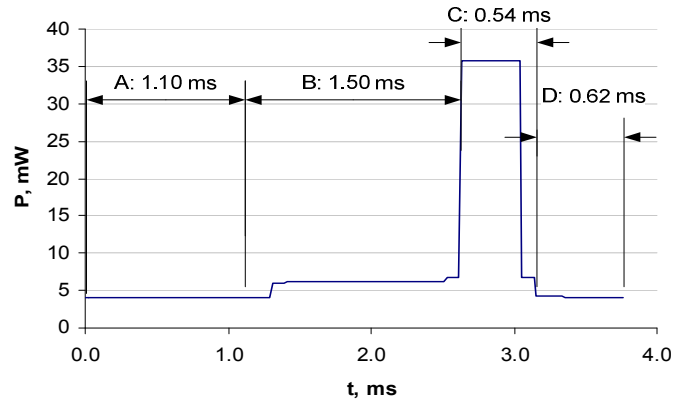


Figure 6. Power consumption during wireless measurement data transmitting: A – transmission packet setup; B – powering up the wireless module; C – actual transmission; D – processing reply from main module

Taking into account measurement and data transmission energy, 0.020 mWs and 0.032 mWs respectively as well as energy consumed during sleep and all time periods, it is possible to calculate the estimated battery service life. Battery service life estimation is made assuming that environment temperature is above 0 °C, because temperature in live beehive is always positive:

$$E_{10min} = E_{Tx} + E_m + T_{sleep} \cdot P_{sleep} =$$

$$= 0.032 + 0.02 + 599.96734 \cdot 0.148 = 88.62 \text{ mWs},$$

$$T_{serv} = \frac{E_{batt}}{E_{10min} \cdot 6} = 20312h \approx 28 \text{ months}$$

where E_{10min} – energy consumption for 10 minutes, mWs;
 E_{Ti} – transmission energy, mWs;
 E_m – measurement energy, mWs;
 T_{sleep} – sleep time during 10 min, s;
 P_{sleep} – average power during sleep, mW;
 T_{serv} – battery service life, h;
 E_{batt} – energy capacity for 3 Wh battery, mWs.

Used system hardware components with approximate unit price are summarized in the table 1.

TABLE I: Summary of used hardware elements with approx. price

Nr.	Name of the component	Quantity (pieces)	Price per piece (EUR)
Central module			
1.	NRF24L01 Wireless Mini Module	1	5.00
2.	A6 GSM module with GPRS	1	8.00
4.	Microcontroller PIC24F	1	4.00
5.	Temperature sensor LMT85DCKT	1	0.50
6.	Printed circuit board	1	10.00
7.	Solar panel 5W	1	15.00
9.	Acid-lead battery 6V 1 Ah	1	10.00
10.	Voltage reference MCP1501-20E	1	1.00
Sensor node			
11.	Microcontroller PIC16F	10	2.00
12.	Printed circuit board	10	2.50
13.	Temperature sensor LMT85DCKT	10	0.50
14.	NRF24L01 Wireless Mini Module	10	5.00
15.	Batteries AAA 1200 mAh	20	1.50
16.	Battery holder 2469	10	1.50
17.	Voltage reference MCP1501-18E	10	1.00

The cost of materials for one such system in author's case was approximately 208.50 EUR.

III. SYSTEM INSTALLATION PROCESS

System installation process should be as simple as possible, because usually beekeepers are not very open to new technologies and mostly not very experienced IT specialists. In authors case user (beekeeper) needs only to do two steps to begin using the monitoring system: pairing each new wireless sensor node to main module and place them into each hive.

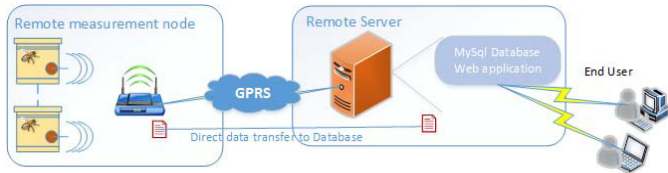


Figure 7. Developed system architecture

In main module SIM card with data plan should be installed to send the data to remote server. System architecture was based on approach described in [17] and is presented in Fig. 7. Still the loss of the GSM signal is critical, because in

that case only data for last 10 minutes would be saved and send to the remote server.

All measurement data, including nodes battery level is stored in MySQL database and available to the user in web application. Application is developed using the Java programming language and has basic functionality, like data reports, charts and additionally it has decision support block, which is used to make any conclusions on bee colony states based on pre-defined algorithms and rules. The implemented rules can be divided in two groups, as described by [12]: individual rules (analysis is based on single colony data) and differential rules (analysis is based on more than one colony (e.g., in one apiary) data). Based on the defined rules, DSS informs the beekeeper if one of the observed states is detected and informs the beekeeper to inspect these colonies. But still the main decision maker remains the beekeeper, therefore it is upon to him to decide, if the provided decision by the system is applicable and should be taken into account or it can be discarded [18].

In addition to the web system it is also possible to develop mobile application or even a smart watch app which will inform beekeepers about changes in the apiary in real time (Fig. 8 and Fig. 9). As well, user alerting about the nodes battery level can be added.

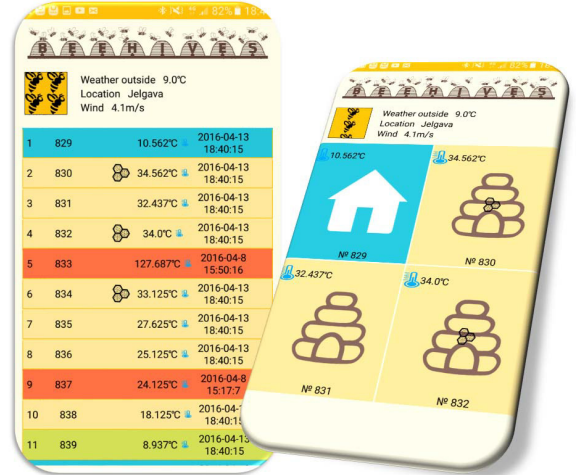


Figure 8. Prototype of application for Android OS



Figure 9. Concept of smart watch application for bee colony monitoring

IV. CONCLUSIONS

Precision Beekeeping direction of the Precision Agriculture is still in developing stage. Nowadays there are different measurement systems available for real-time honeybee colony monitoring. Recently IoT is trending in the field of ICT and represents the future of computing and communication. Implementation of IoT principles in Precision beekeeping could give a significant impact on PB development.

Proposed honey bee monitoring system uses power efficient sensor nodes that consists of low cost, very accurate (± 0.4 °C) temperature sensors with a low power wireless communication modules. Such a system can be set up also in a remote areas, since the main module is energetically self-sufficient and communicates with the database over cell network. Sensor nodes can communicate with the main module in a distance of 20 m in open field.

Data are sent to the main module in a minute interval (containing measurements of temperature and battery level of the specific sensor node) where the main module then performs transmission of those data after 10 minutes. Therefore described system can be considered as a crucial part in a Smart apiary management minimizing the need for manual hive inspection even if the hives are placed in rural areas or forests.

For future discussions and more system energy saving purposes it can be considered to send from wireless nodes only temperature fluctuations values and in case temperature is constant do not send repeating values.

REFERENCES

- [1] U. Warnke, "Bees, birds and mankind," *Destroying Nat. by "electrosmog"*, p. 47, 2009.
- [2] A. Zacepins and T. Karasha, "Application of temperature measurements for the bee colony monitoring: a review," in *Proceedings of the 12th International Scientific Conference "Engineering for Rural Development"*, 2013, pp. 126–131.
- [3] Y. Bo and H. Wang, "The Application of Cloud Computing and The Internet of Things in Agriculture and Forestry," in *Proceedings of the 2011 International Joint Conference on Service Sciences (IJCSS)*, 2011.
- [4] B. Wang, "Review on internet of things," *J. Electron. Meas. Instrum.*, vol. 23, no. 12, pp. 1–7, 2009.
- [5] J. Zhao, J. Zhang, Y. Feng, and J. Guo, "The Study and Application of the IOT Technology in Agriculture," in *Computer Science and Information Technology (ICCSIT)*, 2010, pp. 462–465.
- [6] A. Katasonov, O. Kaykova, O. Khriyenko, S. Nikitin, and V. Terziyan, "Smart Semantic Middleware for the Internet of Things," *Icinco-Icso*, pp. 169–178, 2008.
- [7] L. Yan, Y. Zhang, L. T. Yang, and H. Ning, *The Internet of things: from RFID to the next-generation pervasive networked systems*. 2008.
- [8] D. Brock and E. Schuster, "On the Semantic Web of Things," in *Semantic Days 2006*, 2006.
- [9] N. Wang, N. Zhang, and M. Wang, "Wireless sensors in agriculture and food industry—Recent development and future perspective," *Comput. Electron. Agric.*, vol. 50, no. 1, pp. 1–14, Jan. 2006.
- [10] C. Chong and S. P. Kumar, "Sensor networks: Evolution, opportunities, and challenges," *Proc. IEEE*, vol. 91, no. 8, pp. 1247–1256, 2003.
- [11] A. Zacepins, E. Stalidzans, and J. Meitalovs, "Application of information technologies in precision apiculture," in *Proceedings of the 13th International Conference on Precision Agriculture (ICPA 2012)*, 2012.
- [12] A. Zacepins, V. Brusbardis, J. Meitalovs, and E. Stalidzans, "Challenges in the development of Precision Beekeeping," *Biosyst. Eng.*, vol. 130, pp. 60–71, Feb. 2015.
- [13] A. A. Altun, "Remote Control of the Temperature-Humidity and Climate in the Beehives with Solar-Powered Thermoelectric System," *J. Control Eng. Appl. Informatics*, vol. 14, no. 1, pp. 93–99, 2012.
- [14] G. Hudson and C. Hudson, "Bee Hive Monitor | OpenEnergyMonitor." [Online]. Available: <http://openenergymonitor.org/emon/beemonitor>.
- [15] Carl, "PIC ' n the Beehive." [Online]. Available: <http://kurskelectronik.ru/lifsys/collegs/PIC-the->.
- [16] A. Zacepins, A. Kviesis, E. Stalidzans, M. Liepniece and J. Meitalovs, "Remote detection of the swarming of honey bee colonies by single-point temperature monitoring," in *Biosystems Engineering*, 2016, pp. 76–80.
- [17] A. Kviesis, and A. Zacepins, A. "System Architectures for Real-time Bee Colony Temperature Monitoring" in *Procedia Computer Science*, 2015, pp. 86–94.
- [18] A. Kviesis, A. Zacepins, and G. Riders, "Honey bee colony monitoring with implemented decision support system," in *Engineering for Rural Development*, 2015, pp. 446–451.